

Burrows-Wheeler transformacija i FM index

Projektni zadatak iz predmeta
Genomska informatika

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Burrows-Wheeler transformacija

FM index

i	Rotacije	Sortirane rotacije	F (Prva)	L (Poslednja)	SA	Occ(\$)	Occ(A)	Occ(B)	Occ(N)
0	BANANA\$	\$BANANA	\$	A	6	0	1	0	0
1	ANANA\$B	A\$BANAN	A	N	5	0	1	0	1
2	NANA\$BA	ANA\$BAN	A	N	3	0	1	0	2
3	ANA\$BAN	ANANA\$B	A	B	1	0	1	1	2
4	NA\$BANA	BANANA\$	B	\$	0	1	1	1	2
5	A\$BANAN	NA\$BANA	N	A	4	1	2	1	2
6	\$BANANA	NANA\$BA	N	A	2	1	3	1	2

- Algoritam:
- 1. $L = BWT(T)$
 - 2. First column
 - 3. Suffix Array
 - 4. Occurrence matrix

- Problemi:
- 1. Sporo pretraživanje prethodnog karaktera $O(m)$
 - 2. Čuvanje rankova
 - 3. Određivanje mesta preklapanja
- SA sample

Karakter	Indeks u F koloni
\$	0
A	1
B	4
N	5

Analizirani genomi

Coffea Arabica	Mus Pahari	Ornithorhynchus anatinus (Platypus)
Chromosome 1c	Chromosome X	Chromosome X1
ATGCATG, TCTCTCTA, TTCACTACTCTCA	ATGATG, CTCTCTA, TCACTACTCTCA	GCAGTGG, ACAAAGGGCAAGAAGAA, GATGTTGTGGCAGAGCTTGAAGATT
		

Optimizacije

1) Konstrukcija BWT i SA - divsufsort jedan od najbržih poznatih algoritama

Naivni pristup sa sortiranjem rotacija je $O(n^2 \log n)$, dok je divsufsort:

Worst-case time: $O(n \log n)$

Memory space: $5n + O(1)$ bytes

2) Kontrolne tačke (Checkpointing): Čuva broj karaktera samo u fiksnim intervalima (npr. svaka 128. pozicija). Da bi se pronašao broj na bilo kojoj drugoj poziciji, preuzima najbližu kontrolnu tačku (checkpoint) i izračunava preostali broj.

3) Uzorkovanje sufiksnog niza: Čuva samo svaki k -ti unos sufiksnog niza. Pozicije koje nisu direktno sačuvane rešavaju se korišćenjem koraka unazad sa BWT-om.

Checkpoints i uzorkovanje SA na prostom primeru – BANANA L : A, N, N, B, \$, A, A

i	Occ(\$)	Occ(A)	Occ(B)	Occ(N)
0	0	1	0	0
4	1	1	1	2

Occ('A', 6) = ?

Occ('A', 4) = 1

Occ('A', 6) = Occ('A', 4) + (L[5]=='A') + (L[6]=='A')

Occ('A', 6) = 3

i	Occ(\$)	Occ(A)	Occ(B)	Occ(N)
0	0	1	0	0
1	0	1	0	1
2	0	1	0	2
3	0	1	1	2
4	1	1	1	2
5	1	2	1	2
6	1	3	1	2

SA[3] = ?

i	SA
0	6
1	/
2	/
3	/
4	0
5	/
6	/

L[3] = B

C['B'] = 4, Occ['B', 3] = 1

Korak: 4+1-1 = 4

SA[4] = 0

Broj koraka: 1

SA[3] = SA[4] + broj_koraka = 0 + 1 = 1

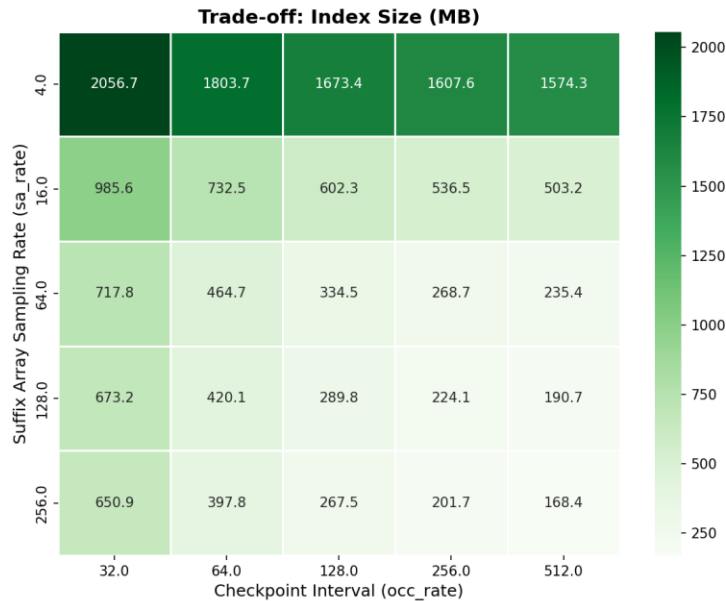
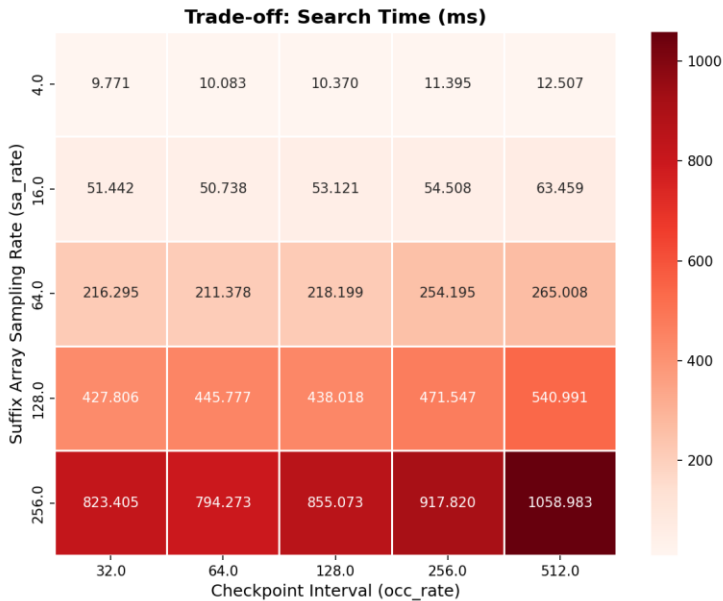
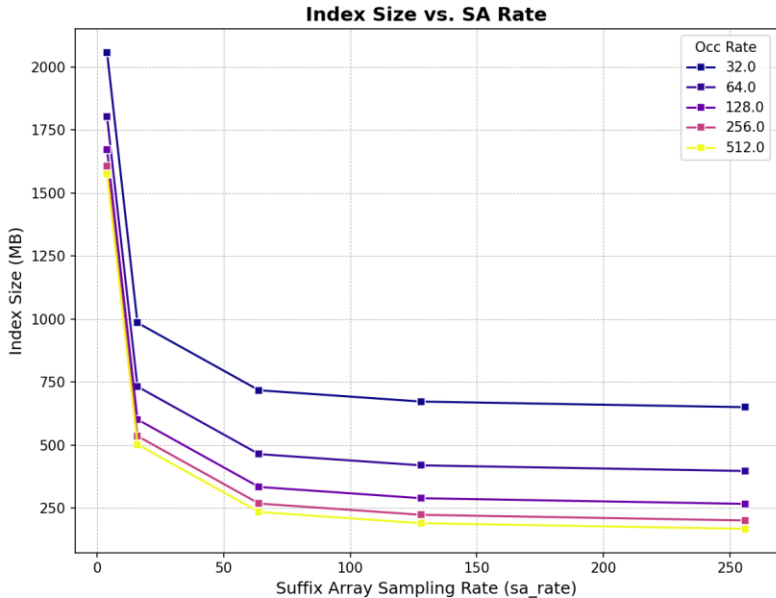
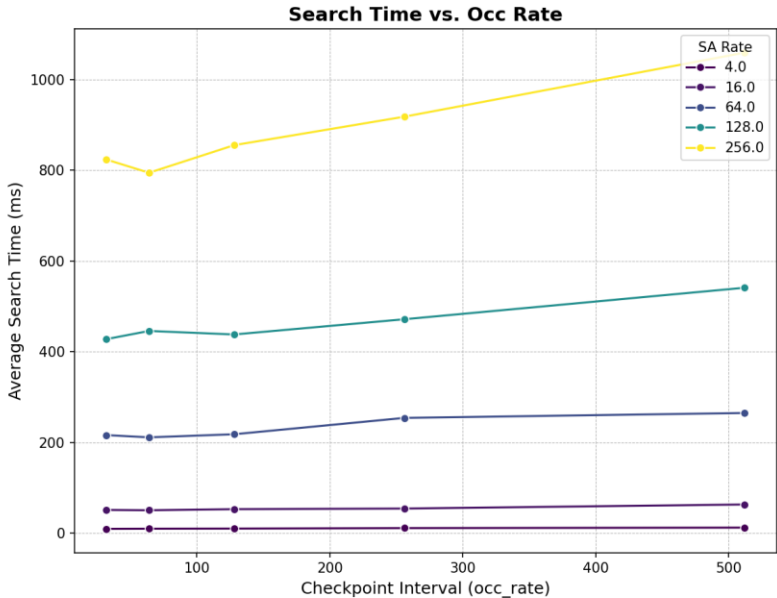
Karakter	Indeks u F koloni
\$	0
A	1
B	4
N	5

i	SA
0	6
1	5
2	3
3	1
4	0
5	4
6	2

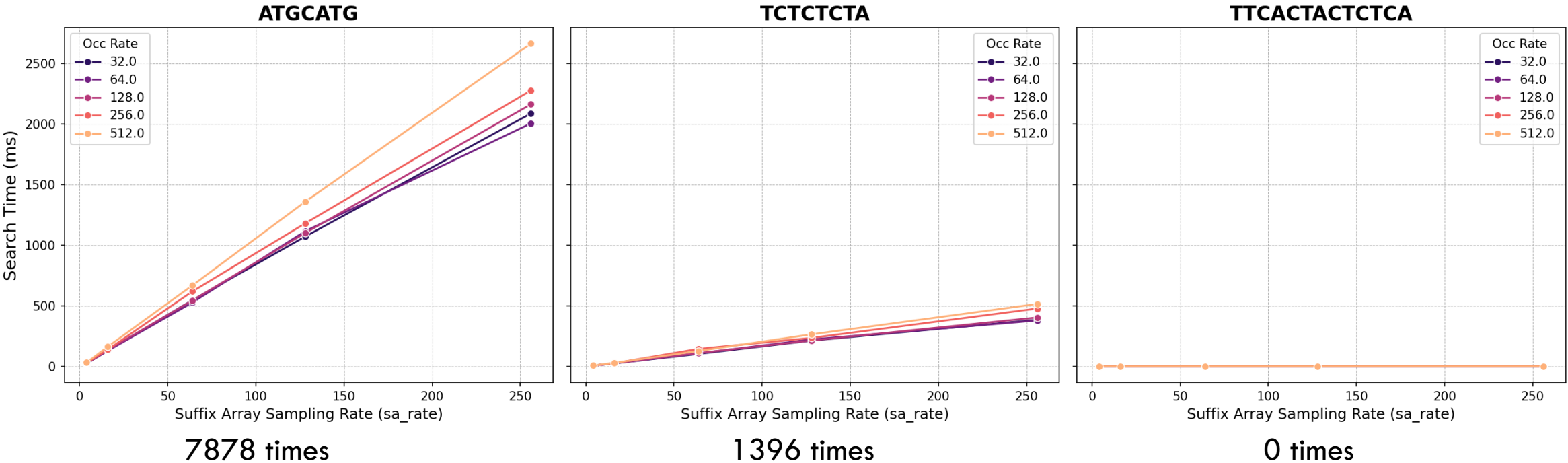
Rezultati



FM-Index Performance Analysis for: Coffea arabica



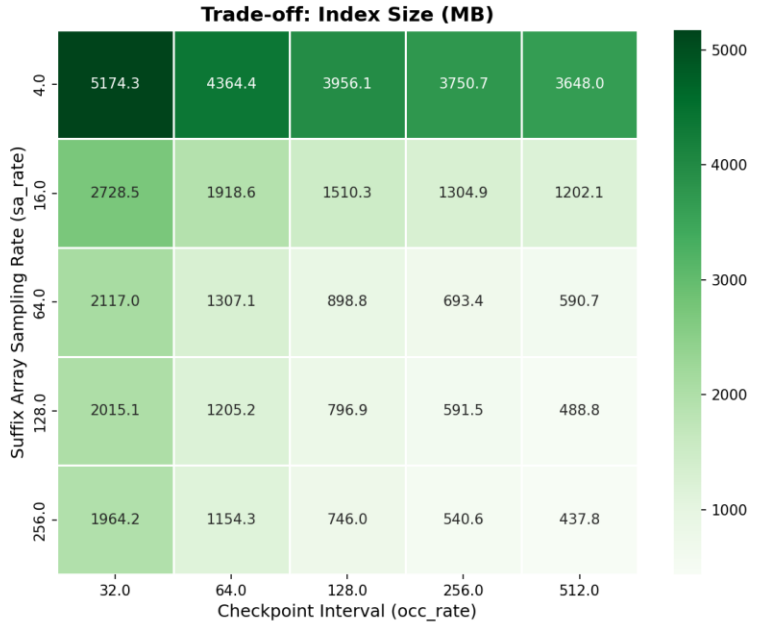
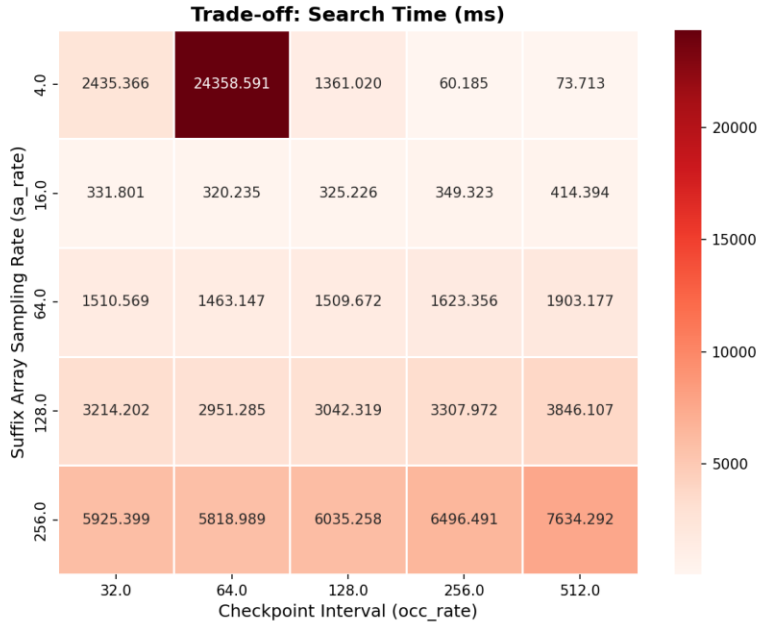
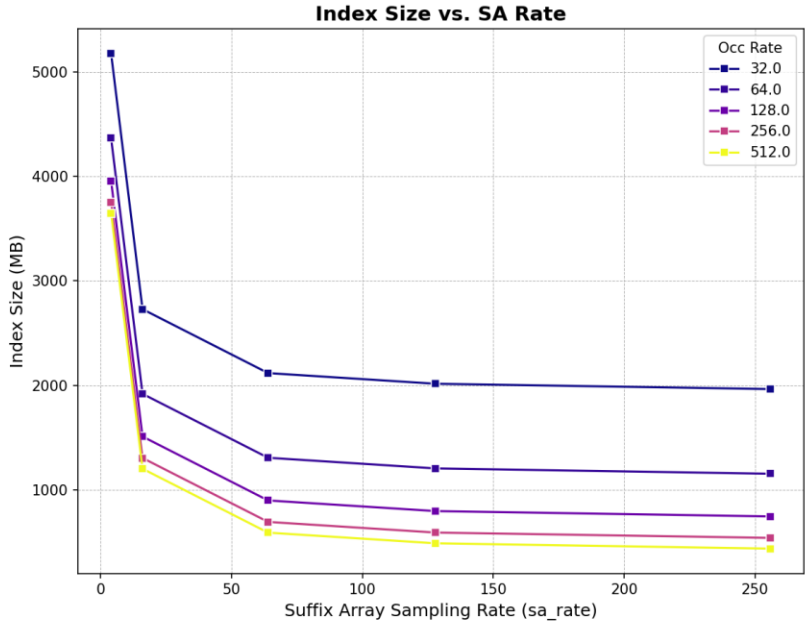
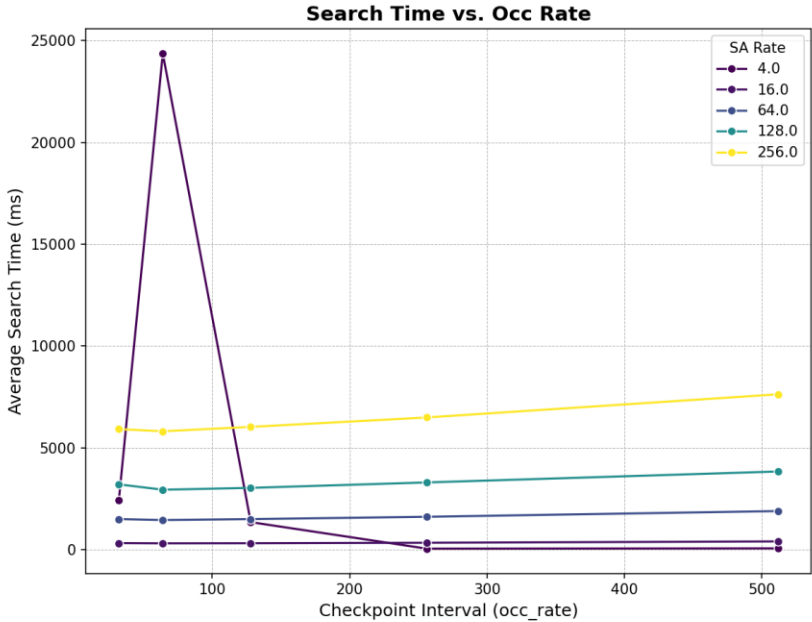
Performance Analysis by Pattern for: Coffea arabica



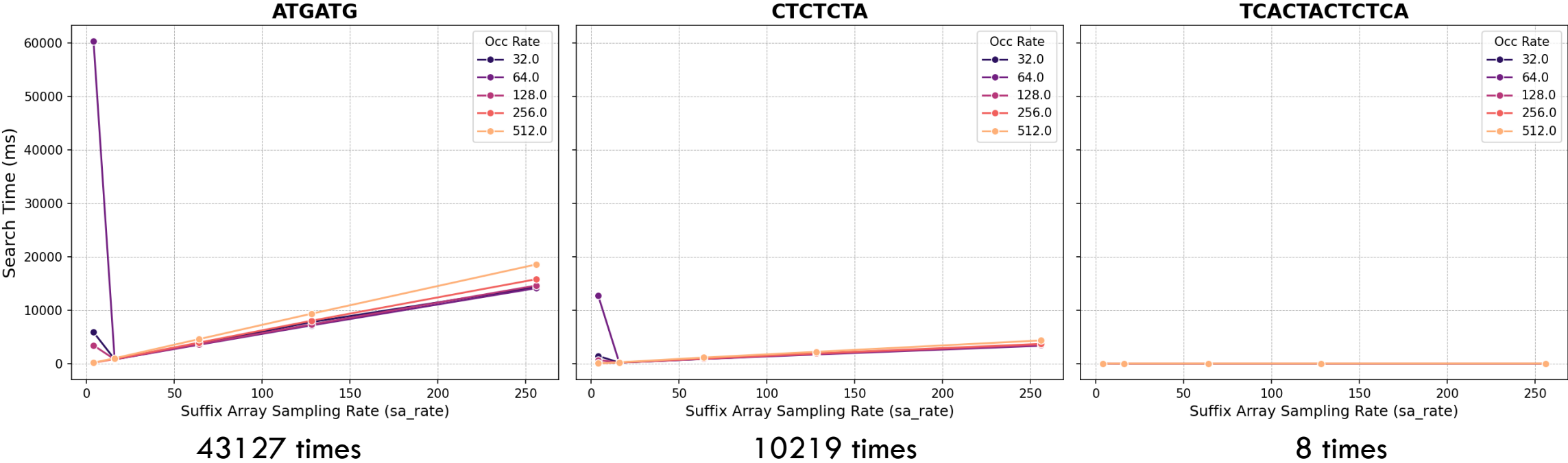
Rezultati



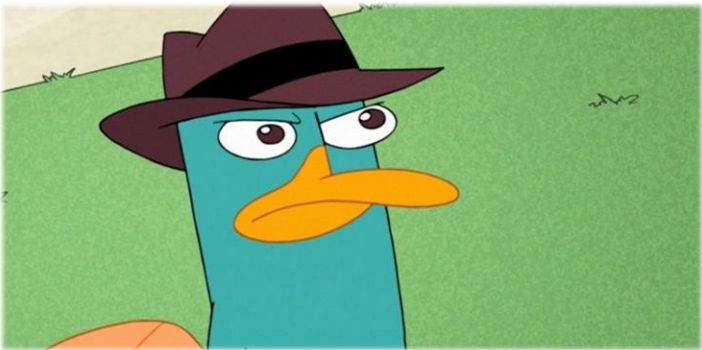
FM-Index Performance Analysis for: Mus pahari



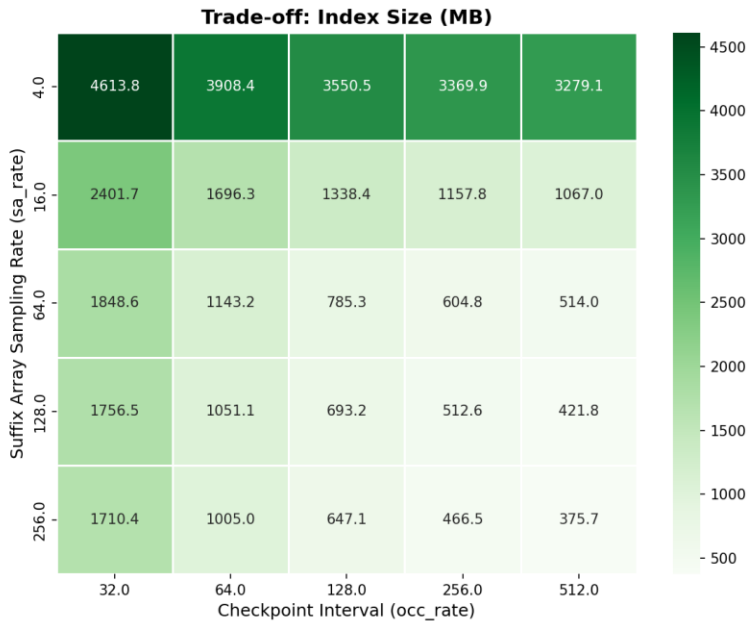
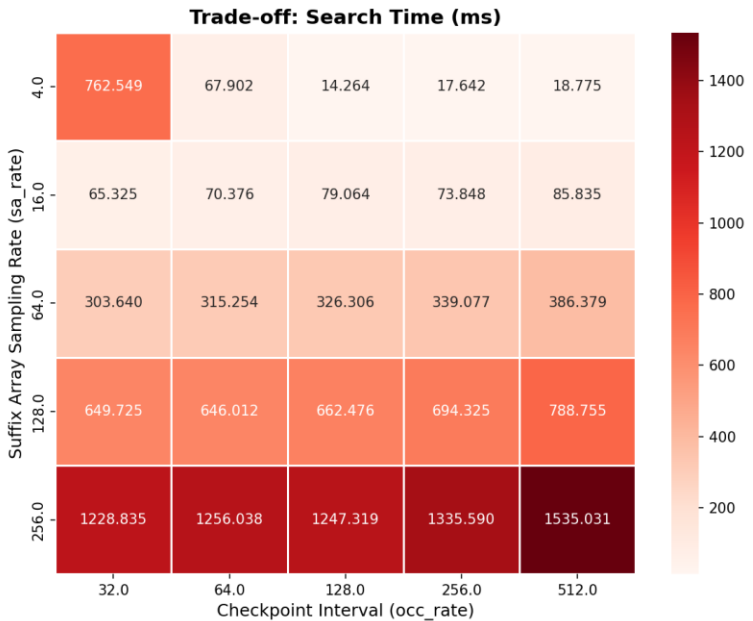
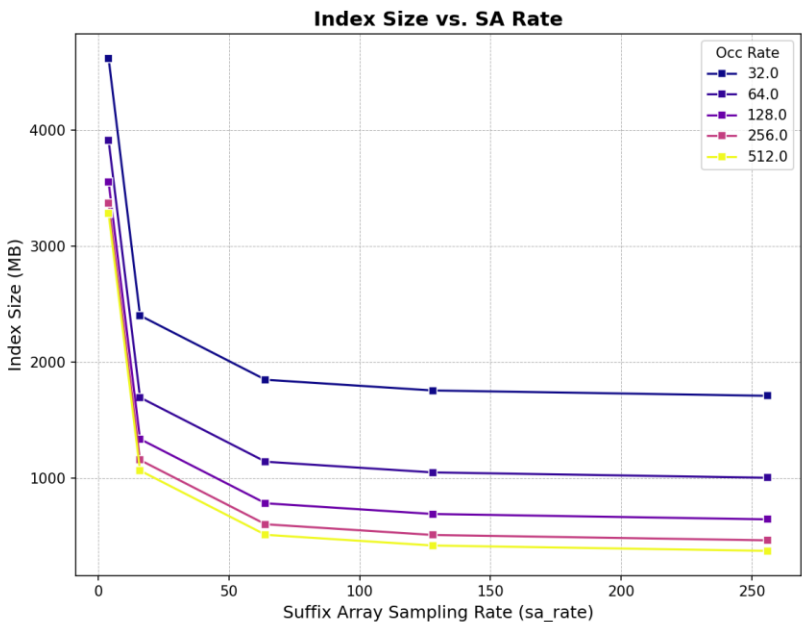
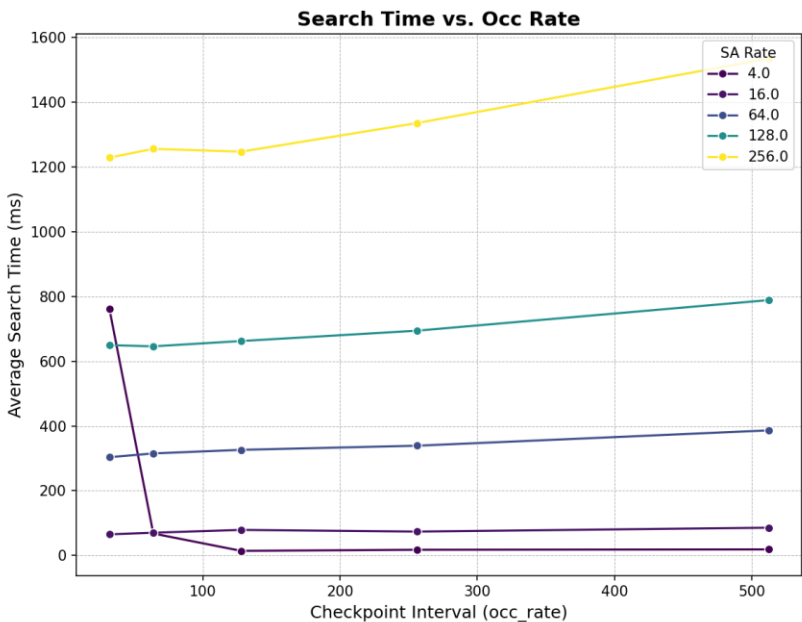
Performance Analysis by Pattern for: Mus pahari



Rezultati



FM-Index Performance Analysis for: Platypus



Performance Analysis by Pattern for: Platypus

